

7. Which argument of `theme()` will help you modify the axis tick labels in R?

`axis.text`



8. What is the function for limiting the range of the x-axis in R?

`xlim()`



9. Which R function projects a portion of Earth, which is approximately spherical, onto a flat 2D plane?

`coord_map`



10. Given the following code in R:

```
p <- ggplot(mtcars, aes(mpg, wt))+geom_point()
```

You want to extend the scatter plot to start with 0. How would you accomplish this?

```
p + expand_limits(x = 0, y = 0)
```



11. Consider the following data in R:

```
library(ggplot2)
```

```
x <- c(5, 8, 6, 2, 1, 5, 4, 2, 3, 7, 8, 4)
```

```
y <- c(1, 2, 6, 4, 8, 4, 6, 1, 7, 5, 2, 9)
```

```
z <- c(5, 4, 4, 5, 3, 2, 2, 1, 5, 4, 2, 1)
```

```
df <- data.frame(x, y, z)
```

How can you create a point plot between `x` and `y`, where `z` is the new dimension with exponential transformation?

```
ggplot(df, aes(x = x, y = y, size = z)) + geom_point(alpha = 0.2) + scale_size_continuous( trans="exp", range=c(1, 5))
```



12. How is scale modification useful in plotting relationships?

Modifying scales can present better relationships of unclear dependencies ✓

13. Which argument holds the names of the base R bar chart categories?

names.arg

✓

14. Given the following code in R:

```
p <- ggplot(diamonds, aes(carat, price))+  
geom_point()
```

How would you add x- and y-axis labels, and a title to the plot?

```
p + labs(x = 'Carat of the diamond', y = 'Price of the diamond', title = 'Carat Vs Price')
```



15. Using the iris data set in R, you create a scatter plot between `Sepal.Width` and `Petal.Width`. How can you identify different species with different colors?

```
ggplot(iris, aes(x = Sepal.Width, y = Petal.Width, col = Species)) +  
geom_point()
```



16. You are working with the `mtcars` dataset in R. You are plotting a scatter plot with the following code:

```
library(ggplot2)
p <- ggplot(mtcars, aes(wt , mpg))
+geom_point()
```

How would you add a layer of `facet_grid` and create a facet in rows for `'cyl'`?

```
p + facet_grid(cyl ~.)
```



17. Which factor can be deduced from a heatmap of five vectors?

Correlation ✓

18. The `mtcars` dataset in R contains 11 columns of different features of cars, all of numeric type. How would you create a plot that shows the relationship between `mpg` (mileage per gallon) and `hp` (horse power), with the different numbers of `cyl` (cylinders) in different colors?

```
ggplot(mtcars, aes(x = mpg, y = hp, col = as.factor(cyl))) +  
  geom_point()
```



19. Given the following code in R:

```
p <- ggplot(mpg, aes(displ, hwy))+ geom_point()
```

`p` is a scatter plot between `displ` and `hwy`. How would you plot subplots for a class variable with unique ranges for each plot?

```
p+  
facet_wrap(~class, scales = "free")
```



1. Consider the following information in R:

```
library(ggplot2)
df <- data.frame(category = c('4th', '3rd', '2nd', '1st'),
                  n = c(885, 706, 285, 325),
                  per = c(40.2, 32.1, 12.9, 14.8),
                  pos = c(20.1, 56.25, 78.75, 92.6),
                  cols = c("#0073C2FF", "#EFC000FF", "#868686FF", "#CD534CFF"))
```

Code 1

```
ggplot(df, aes(x = 2, y = per, fill = category)) +
  geom_bar(stat = "identity", color = "white") +
  coord_polar(theta = "y", start = 0)+
  geom_text(aes(y = pos, label = per), color = "white") +
  scale_fill_manual(values = mycols) +
  theme_void()+
  xlim(0.5, 2.5)
```

Code 2

```
ggplot(df, aes(x = 2, y = per, fill = category)) +
  geom_bar(stat = "identity", color = "white") +
  coord_polar(theta = "y", start = 0) +
  annotate('text', x = 1, y = 0, label = 'c(40.2, 32.1, 12.9, 14.8)') +
  scale_fill_manual(values = mycols) +
  theme_void()+
  xlim(0.5, 2.5)
```

How does the creation of a donut chart differ when executing the two codes?

Code 1 labels each group, whereas code 2 labels the center of the donut chart



2. Which element function can you use to modify tick marks on the axes in R?

`element_line()`



3. In which layer of ggplot2 are colors applied in a plot?

Aesthetic ✓

4. Given the code in R:

```
ggplot(iris, aes(x = Sepal.Length, y = Petal.Length, col = 'blue')) + geom_point()
```

How would you change the code to plot a scatter plot with blue points?

```
ggplot(iris, aes(x = Sepal.Length, y = Petal.Length)) +  
  geom_point(color = 'blue')
```



5. In R, what value is passed to the argument `stat` to override the default connection between `geom_bar()` and `stat_count()` ?

identity



6. Given the following code in R:

```
p <- ggplot(diamonds, aes(price, carat))+geom_point()
```

This code in R describes a plot showing the relationship between `price` and `carat`. How would you plot the same relationship while comparing `clarity`?

```
p +facet_wrap(~clarity)
```



7. You are working with the `mtcars` dataset in R. What will tell you the mean of `wt` and `mpg`?

```
mtcars %>% summarise(mean_mpg = mean(mpg), mean_wt = mean(wt))
```



8. Given the following code in R:

```
p <- ggplot(msleep, aes(sleep_total, bodywt)) +  
  geom_point()
```

How would you change the color of the x-axis tick labels to red?

```
p + theme(axis.text.x = element_text(colour = "red"))
```



7. Consider the following code in R:

```
library(ggplot2)

df <- data.frame(category = c('A', 'B', 'C'),
                  max_val = c(0.1, 0.4, 0.1),
                  min_val = c(0.0, 0.5, 0.4))

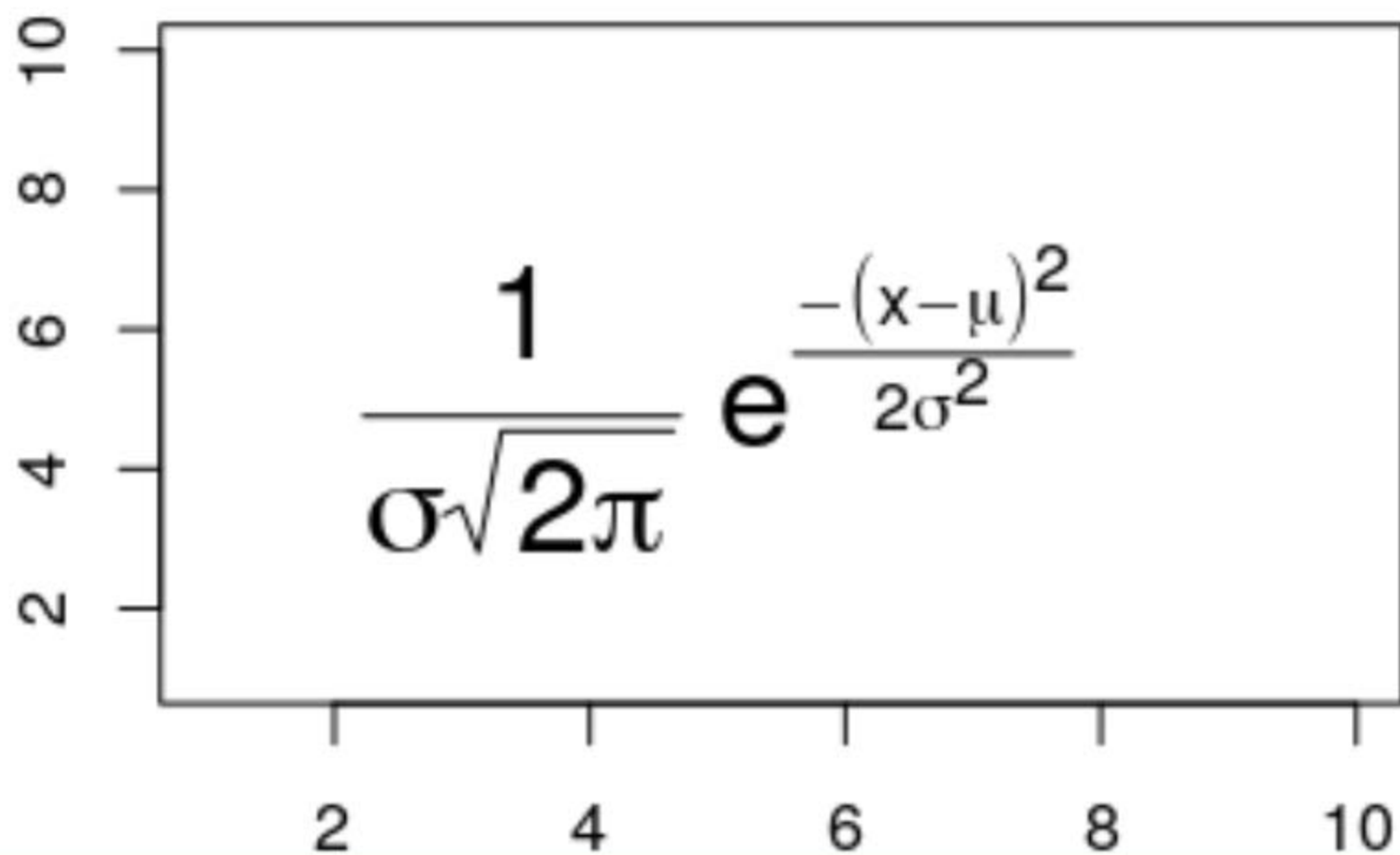
# Make the plot
ggplot(df, aes(P, xmax = 4, xmin = 3)) +
  geom_rect() +
  coord_polar(theta="y") +
  xlim(c(0, 4)) +
  theme(panel.grid=element_blank()) +
  theme(axis.text=element_blank()) +
  theme(axis.ticks=element_blank()) +
  labs(title="")
```

What would the value of `P` be in the code if you want to build a donut chart using the `geom_rect` function?

`fill = category, ymax = max_val, ymin = min_val`



8. You are working with R base graphics. What should be the value of `labels` to produce a mathematical expression as shown in the figure?



```
expression(paste(frac(1, sigma*sqrt(2*pi)), " ", plain(e)^{frac(-(x-mu)^2, 2*sigma^2)}))
```



9. Which R function projects a portion of Earth, which is approximately spherical, onto a flat 2D plane?

`coord_map`



10. Consider the following data in R:

```
library(ggplot2)

x <- c(5, 8, 6, 2, 1, 5, 4, 2, 3, 7, 8, 4)
y <- c(1, 2, 6, 4, 8, 4, 6, 1, 7, 5, 2, 9)
z <- c(5, 4, 4, 5, 3, 2, 2, 1, 5, 4, 2, 1)

df <- data.frame(x, y, z)
```

How can you create a point plot between `x` and `y`, where `z` is the new dimension with exponential transformation?

```
ggplot(df, aes(x = x, y = y, size = z)) + geom_point(alpha = 0.2) + scale_size_continuous( trans="exp", range=c(1, 5))
```



11. You are working with the animals dataset from the MASS package in R. After analyzing the data, you find a linear relationship between `log()` of body weight and brain weight. How would you change the code below to scale the x- and y-axes to show a linear relationship?

```
ggplot(Animals, aes(brain, body))+ geom_point()
```

```
ggplot(Animals, aes(brain, body))+  
  geom_point()+  
  scale_x_log10()+  
  scale_y_log10()
```



12. Given the following code in R:

```
p <- ggplot(diamonds, aes(carat, price))+  
  geom_point()+  
  xlab("Carat")+  
  ylab("Price")
```

In this ggplot object, you are plotting a scatter plot. How would you plot the same scatter plot without any labels and without changing the initial object `p` ?

```
p + labs(x = NULL, y = NULL)
```



13. What is the benefit of using `facet_wrap` in R?

It provides better use of space



14. What is true about choropleth maps in R?

They give the impression of uniform values within each sub-area



15. How would you create a 3D scatter plot in R between `Sepal.Length`, `Sepal.Width`, and `Petal.Length` of the iris dataset without a dividing surface?

```
scatter3d(x = iris$Sepal.Length, y = iris$Sepal.Width, z = iris$Petal.Length, surface = F)
```



16. Consider the following labels in R

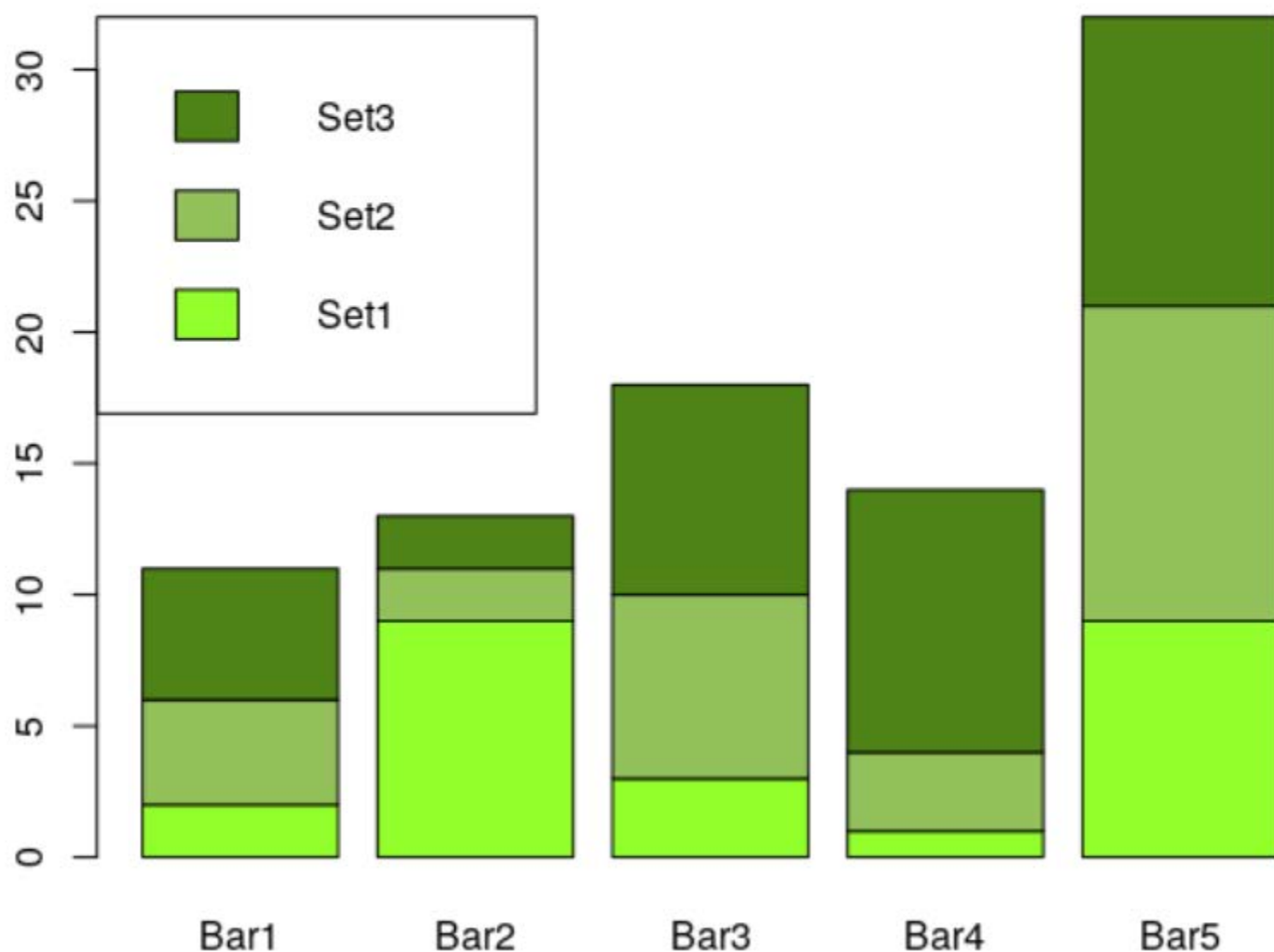
```
colors <- c("#92FF03", "#93C156", "#4E8309")
bars <- c("Bar1", "Bar2", "Bar3", "Bar4", "Bar5")
sets <- c("Set1", "Set2", "Set3")
```

Additionally, 15 values are provided that are stored in a variable `values` :

```
2,  9,  3,  1,  9
4,  2,  7,  3, 12
5,  2,  8, 10, 11
```

Using the above information and assuming the `values` variable is stored in a valid data structure, which code best describes the formation of a stacked bar chart as shown in the figure?

Stacked bars



```
barplot(values, main = 'Stacked bars', names.arg = bars,
        col = colors, legend.text = sets, args.legend = list(x = 'topleft'))
```



17. You have created a correlogram with the mtcars dataset in R:

```
library(corrgram)
corrgram(mtcars, order=NULL, lower.panel=panel.shade, upper.panel=NULL, text.panel=panel.txt, main="Correlogram")
```

Which code would produce a correlogram that is ordered, with a confidence ellipse in the upper panel and a magnitude pie in the lower panel?

```
corrgram(mtcars, order=TRUE, lower.panel=panel.pie, upper.panel=panel.ellipse, text.panel=panel.txt, main="Correlogram")
```

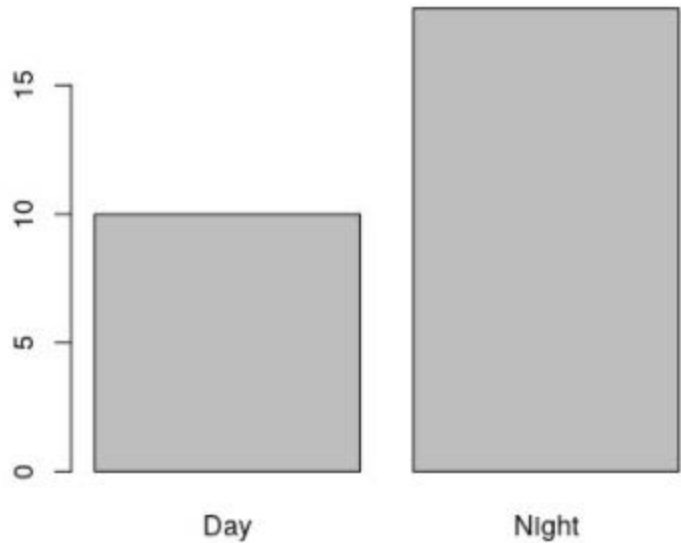


18. Given a dataframe:

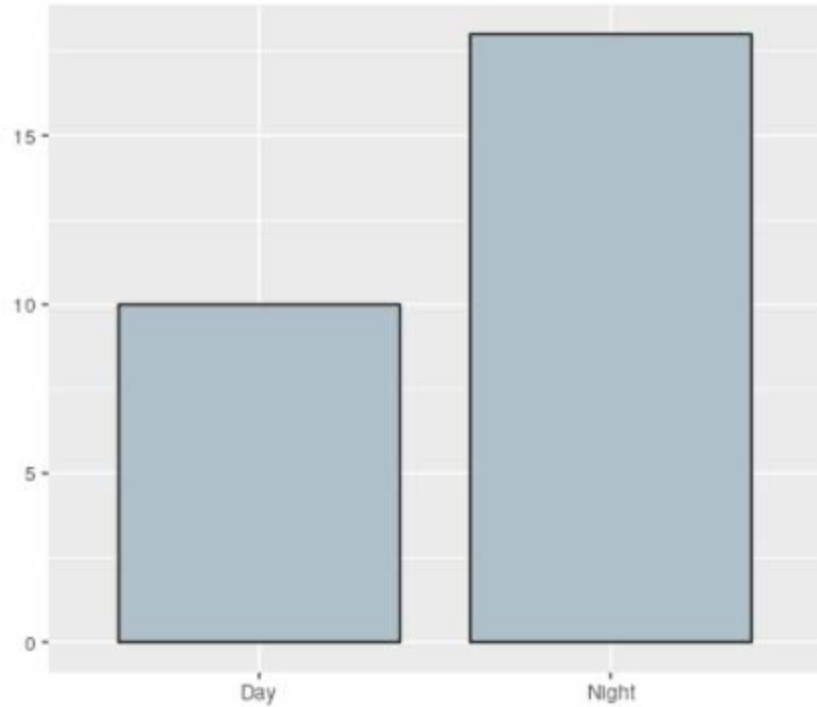
```
df <- data.frame(  
  time = factor(c("Day","Night"), levels=c("Day","Night")),  
  average_bill = c(10, 18)  
)
```

How many function calls are needed by a base R plot and a ggplot to build a bar chart as shown in the figure

Built using base R



Built using ggplot2



base R: 1
ggplot2: 4

